

Designing a user centric VR UI/UX for scientific visualisation

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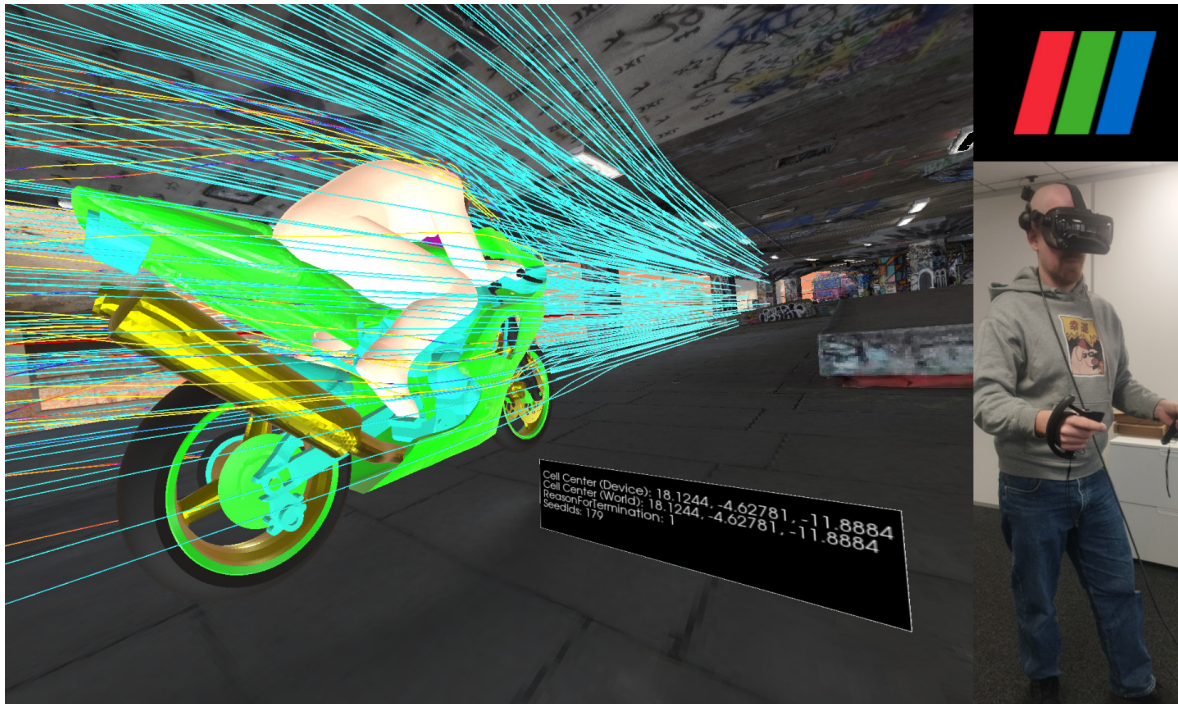


Figure 1: Visualization of the OpenFOAM motorbike in a Virtual Reality environment in ParaView

ABSTRACT

Scientific visualization (scivis) is an important tool for all scientific domains that aims to reveal the most salient information out of raw data and display it in an intuitive way so that users can make decisions based on their data characteristics. Virtual reality (VR) is a dedicated technique to render three-dimensional data as a real world interactive and immersive scene. ParaView [2] is a leading open-source scivis software providing many tools for data analysis and visualization. ParaView has been providing VR support for some time now but it has been improved greatly in the last few years, based on feedback from users in the industry and academia alike, especially in regards to menu interactions and collaboration. This work brings insight into the new VR interface and how it provides clear and intuitive access to features and how collaborative VR can be an integral tool for scivis in the future.

Index Terms: ParaView, Virtual Reality, Collaboration, User interface.

1 INTRODUCTION

Ever since the breakout years of virtual reality, in the 90s, scientific visualisation in VR has always been considered, studied and

implemented in many different ways. The theoretical groundwork of what scientific visualization in virtual reality would look like is done [3], which includes vocabulary and concepts, as well as suggested standards for the industry and the academy to follow, however, no implementation is provided. The different concepts are refined and a custom implementation of virtual reality for scientific visualization is shared [10]. With this first implementation, hardware and software challenges are highlighted by this article. A complete status of different VR implementations of scientific visualization is made [7] as well as studying different usecases and application. While a few generic implementations are listed, ParaView in particular, most of the implementations are specific to their field or focused on learning and education. Finally in recent journal issue, we see that most papers focus on learning and training, with Unity based specific implementations, with only a few sharing code [1].

So, in 2024, there are a few generic scientific visualization softwares that support virtual reality in both commercial and open source software, and while the hardware and game development software are in constant development, we see more efforts put into custom unity based research prototypes than into generic tools that could benefit the whole community.

While ParaView is a functional solution to get VR working with scientific visualisation, there are many improvements possible in order to make it the de facto solution for any problems related to scivis and virtual reality and let users avoid implementing their own solutions instead. Chhim already presented the status in detail in a poster [5] at PacificVis conference in 2022.

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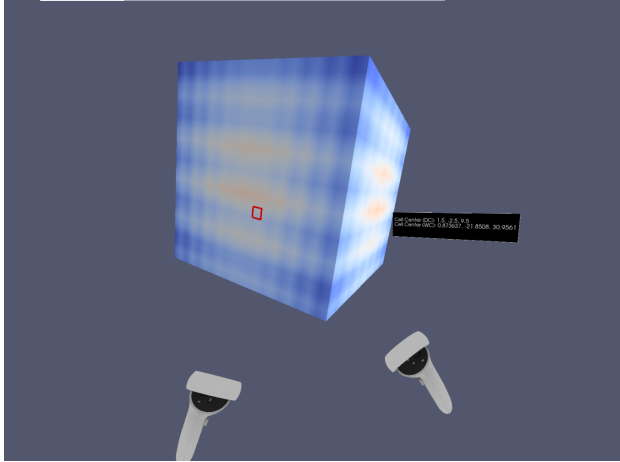


Figure 2: A typical rendering of ParaView with legacy OpenVR plugin.

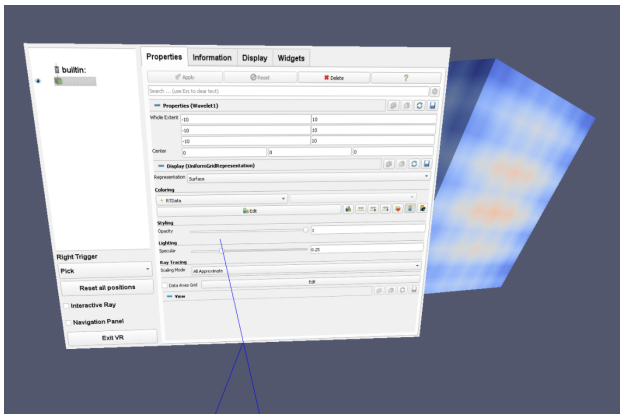


Figure 3: Example of an interaction with the legacy OpenVR menus.

2 PARAVIEW STATUS BEFORE IMPROVEMENTS

As Shermann and Chhim pointed out, ParaView has been providing a VR interface for scientific visualisation for a very long time, through the OpenVR [9] plugin, it was however very much bare-bones and limited in terms of rendering possibilities and most importantly, interactions possibilities. Indeed, the ParaView OpenVR plugin basically copies the ParaView desktop experience into a Virtual Reality space with a few dedicated interactions on top of it.

This figure Fig. 2 shows that the desktop scene is being ported to the VR environment as is. It works but offers little flexibility.

In the VR menu Fig. 3, we used to be able to find the classical pipeline browser and properties panel from ParaView desktop, with no thoughts given to the actual interactivity of the different elements. A few custom options have been added for VR customization but there was no logic nor care given to these so that the VR users could easily interact with it.

Finally, if OpenVR was indeed a pioneer in its time, it is currently being replaced by the OpenXR Khronos standard [8].

In that context a complete redesign was needed, first to offer more customizations and options to VR users, second to provide a more VR-centric UI and UX, finally to support modern standards like OpenXR.

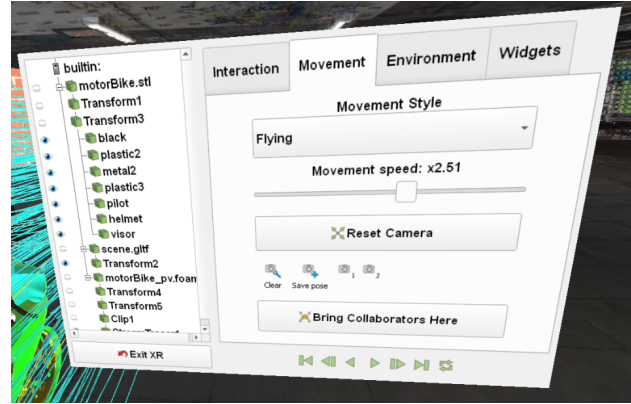


Figure 4: Example of an interaction with the new OpenXR menus.

3 DESIGN AND IMPLEMENTATION OF THE IMPROVEMENTS

In the recent years, numerous contributions have been made to the virtual reality and augmented reality layers of VTK and ParaView. These improvements have been made through the new “XRInterface” plugin as well as through the scientific visualization library VTK upon which ParaView in VR. It lets users control rendering and interaction options. Following this analysis, it was decided to rework the menu deeply in order to improve user experience, a few iterations have been made, using the following pointers. First, we removed obsolete and less-used options. Some of them were non-functional, redundant and experimental. Second, we unified the button in a coherent way, and tried to avoid checkboxes and instead used pushbuttons that were much easier to interact with in VR. Finally we organised the option by category, not only for easier discoverability but also to lighten up the UI Fig. 4.

4 COLLABORATION RELATED IMPROVEMENTS

One of the main interests of VR is collaboration, as in being able to be inside the same scene as other users. ParaView has been supporting collaboration in VR as an experimental feature for a long time, however it was recently improved and industrialized so that anyone with a VR headset can use it easily. First and foremost, the collaboration server [6] is now provided in the binary release of ParaView, for anyone to use and host their own collaborative server. Second, Kitware provides access to a demo server for anyone to test the collaborative feature directly on a dedicated URL, <http://vrserver.kitware.com/>. To use it, two users can just use a simple configuration like this and start collaborating right away Fig. 5. Finally collaboration is now supported across VR and CAVE at the same time, CAVE being using ParaView in an immersive environment which is a room with many screens on different walls, sometimes supporting 3D rendering.

5 FUTURE POSSIBLE IMPROVEMENTS

While these improvements have been released in ParaView, we do not see a large increase in usage of ParaView VR, despite increased communication efforts [5] [4], we believe that ParaView VR could be improved much further, here are a few ideas that we hope to be able to implement in the coming months.

For starters, it is not easy to position oneself inside of the data when the data is of very different scale. There are simple controls

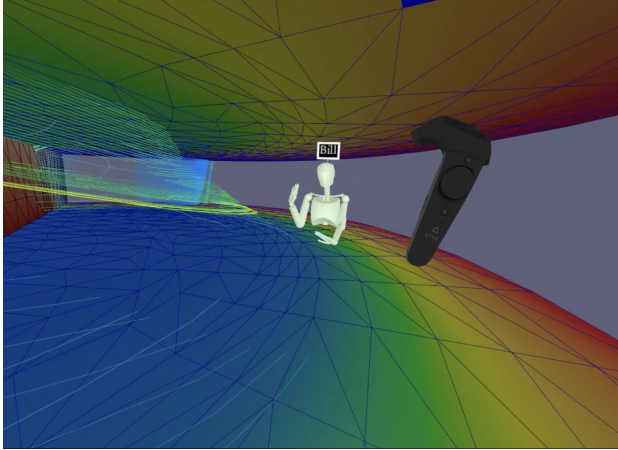


Figure 5: An illustration of collaboration in ParaView.

to modify the scales but this should be handled in a way to make this trivial.

Animation performance is not working as well as one would hope, it is lagging a bit in the VR headset, which can cause nausea. Rendering optimization should be considered here. Also the view that shows what the VR user sees is not working during animation.

Volume rendering is also not performing well. This visualization technique is indeed more heavy than classic surface rendering but it is also essential for scivis, so optimization should be done so this works flawlessly.

Testing features in VR is very hard and there are not many frameworks available to implement it. ParaView currently does not test any of its VR features automatically but rely instead on manual testing, this should be improved.

Data scalability also comes to mind. ParaView has been designed with data scalability in mind, but this does not translate well in the virtual reality support yet. Being able to run smoothly a distributed data processing pipeline while visualizing the result in a VR headset would be a game changer for VR adoption.

Finally, we should consider adding more headsets support. Indeed, each headset comes with its own controls and requires some small but needed work to be supported, and this has not been done for many headsets yet.

6 CONCLUSION

In this article, we presented the previous status of virtual reality support in ParaView and the many improvements that have recently made, especially in regards to user interface and collaboration. We have highlighted how designing for virtual reality is required in order to achieve a good user experience in that context. We suggested many other aspects in which ParaView could be improved in order to becoming the de-facto tool for scientific visualization in virtual reality, especially in regard to user interface.

ACKNOWLEDGMENTS

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